National University of Computer and Emerging Sciences, Karachi
FAST School of Computing
Final Examination, Fall 2021
January 10, 2022, 9:00 am to 12:00 Noon

Course Code: CS3001	Course Name: Computer Networks
Instructor Names: Dr. Aqsa Aslam and Mr. Shoaib Raza	
Student Roll No:	Section No:

Instructions:

- Return the question paper along with the answer script.
- **All questions must be answered in the answer script and according to the sequence given in the question paper. There are 9 questions on 3 pages.**
- In case of any ambiguity, you may make an assumption, however, your assumption should not contradict any statement in the question paper.

Time: 180 minutes **Max Points**: 100 points

Question 1: **[5x3=15 points]**

Final Year students were giving a demo of their e-commerce Website to their supervisor. Their Web Page has 10 different objects. Supervisor asked the following questions during VIVA VOICE.

a) How many RRT will be required to download those objects if they have used the Persistent and Non-Persistent version of HTTP?
b) What is Head of Line (HOL) blocking in the HTTP/2 version and how it was mitigated?
c) How HTTP/3 has increased the performance of Web browsers?
d) Is HTTP a client-server or a peer-to-peer paradigm? Explain your answer.
e) How this e-commerce website can keep browsing trends of its customers?

Question 2: **[5x2=10 points]**

Suppose a user Karim enters the university with his laptop. He is in room S2 in the CS block. Assume that the IT department of the University is running various services to support network operations e.g. DHCP, DNS, File storage, and web application (e.g. Flex). Describe in steps how Karim's laptop:

a) Connects to the Internet.
b) Initiates a query to flex.nu.edu.pk.
c) Connects to another student Talha sitting in the same room (S2).
d) Connects to another instructor Farhan in the EE department.
e) Suppose accessing the File storage has become a bottleneck due to multiple users trying to access the hosted CN video lectures. What application-level solution will you propose to overcome this problem?

Question 3: **[2x5=10 points]**

a) Why IP fragmentation was implemented in IPv4? Explain how IPv6 solves the issue for which IP fragmentation was designed in IPv4.
b) ISPs are slowly changing their IP networks to IPv6. This means they have to interconnect between groups of routers running IPv4 with other groups running IPv6. Suppose a scenario shown in figure 1, where IPv6 packets need to traverse an IPv4 tunnel. How this tunnel is practically implemented?

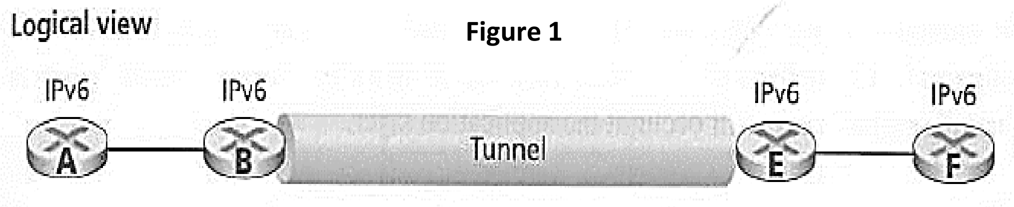

Figure 1

Question 4: **[2x5=10 points]**

a) TCP connection has currently estimated RTT of 30 ms with a deviation of 3 ms. What is the value of the retransmission timer after the next acknowledgement comes in after 28 ms?
b) Does UDP require a mechanism to estimate the RTT between sender and receiver? Briefly explain your answer.

Question 5: **[2x5=10 points]**

Consider figure 2, in which a single router is transmitting packets, each of length L bits, over a single link with transmission rate R Mbps to another router at the other end of the link. Suppose that the packet length is L= 8000 bits and that the link transmission rate along the link to the router on the right is R = 100 Mbps.

a) What is the transmission delay?
b) Assume the average rate is 30. What will be traffic intensity?

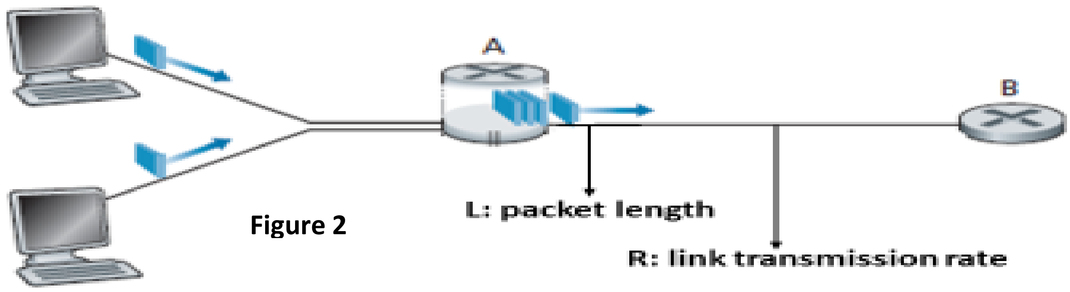

Figure 2

Question 6: **[2x5 =10 points]**

Software-defined networking technology is an approach to network management that enables dynamic and programmatically efficient network. Now answer the following questions:

a) What are four key characteristics of an SDN architecture?
b) How SDN controller makes use of different network applications to provide link state routing in the SND setup? Explain it in steps.

Question 7: **[1x10=10 points]**

Consider the network shown in figure 3. With the indicated link costs, use Dijkstra's shortest-path algorithm to compute the shortest path from *w* to all network nodes. Show how the algorithm works by computing a table. State of Initial routing table is shown below.

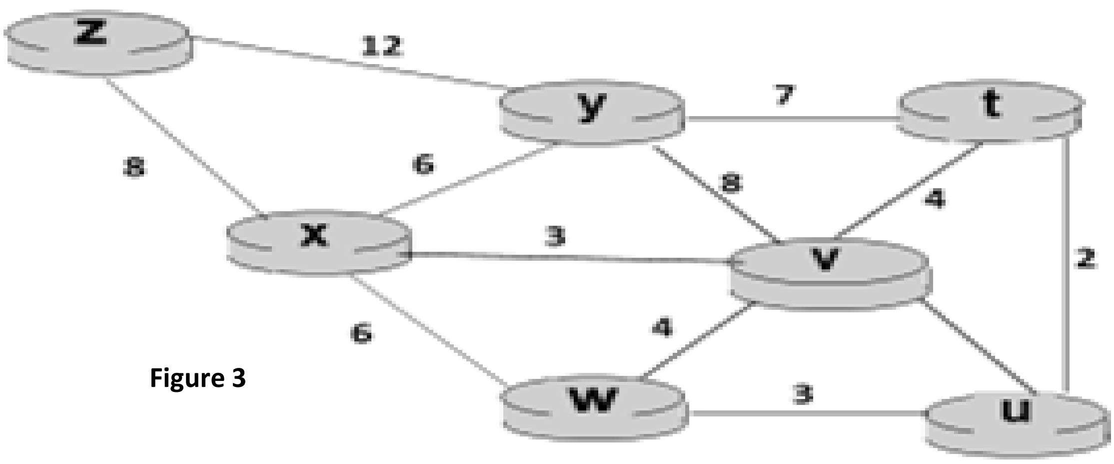

Figure 3

Steps	Node (N')	D(x), p(x)	D(u), p(u)	D(v), p(v)	D(t), p(t)	D(y), p(y)	D(z), p(z)
0							

Question 8: **[5x3=15 points]**

You have been given the IP address 192.168.9.0/24.Now you have to assign IP addresses to the different department's LAN as shown in network figure 4. The network has the following addressing requirements.

a) The Department-1 LAN-1 will require 50 host IP addresses.

b) The Department-1 LAN-2 will require 120 host IP addresses.

c) The Department-2 LAN 1will require 10 host IP addresses.

d) The Department-2 LAN 2will require 18 host IP addresses.

e) The MD LAN-1 will require 28 host IP addresses.

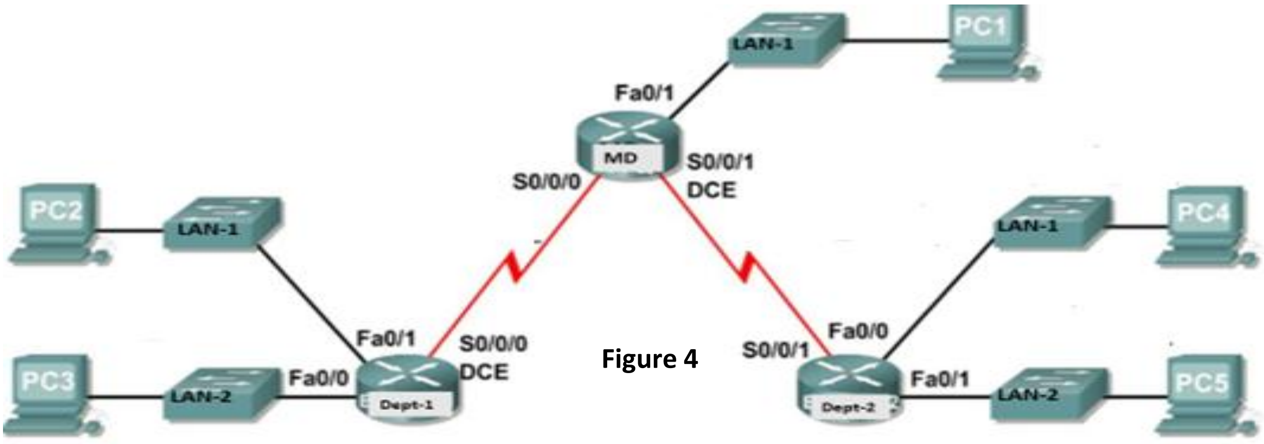

Figure 4

Determine the following.

- Number of Bit Borrowed
- Total Number of Subnets
- Total Number of Host Addresses
- Number of Usable Addresses
- Custom Subnet Mask
- Subnet Range(Network & Broadcast address)

Question 9: **[3+3+4=10 points]**

Consider figure 5 and answer the following questions:

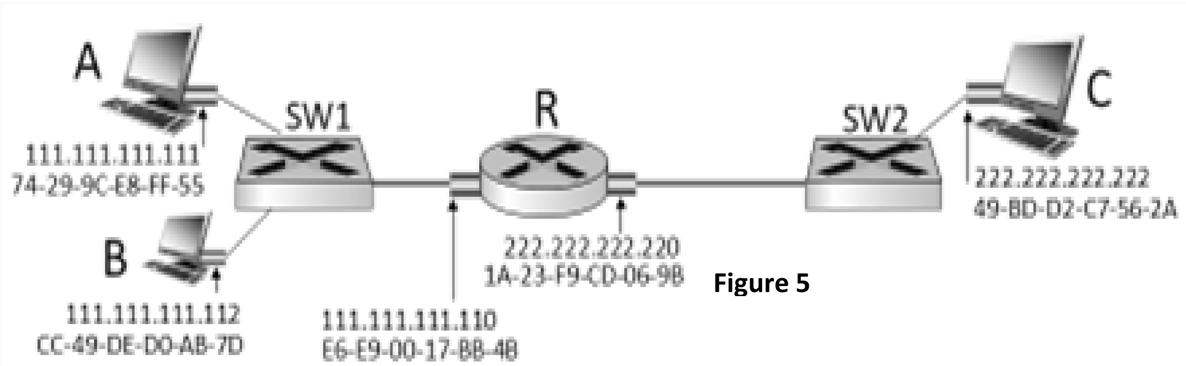

Figure 5

a) Consider sending an IP datagram from PC-A to PC-B. PC- A issues an ARP request because it needs to send a packet to PC- B. In this scenario, what will happen next?

b) Suppose PC- A would like to send IP datagram to PC - C on a remote LAN, however there are currently no mappings in the ARP cache. How will the device obtain a destination MAC address?

c) What will be the content of the destination IP-address field in the header of the IP datagram sent by PC- A to PC-C?

Best of Luck!